TEXAS
The University of Texas at Austin

LLM PROMPTING – RAG

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Outline

- Introduction to RAG
- Semantic Similarity & Database search
- Building a RAG

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RAG

- One downside of generative LLMs is that specific domain knowledge is tricky
  - LLMs learn sentence composition rules
  - LLMs don't learn what part is needed to fix a tractor

- Retrieval Augmented Generation (RAG) is one common solution

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RAG

- We can search a tractor repair manual for information relevant to a question!
  - Retrieval
- We can incorporate the relevant parts (Augmentation) of the manual in an LLM prompt
  - Generation

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RAG

1. Ask a domain specific question
2. Search a database of domain specific information
3. Find the most relevant parts of the database
4. Design a prompt to an LLM that has the question and results from step 3

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RAG

Answer the following question with the context below.

Question: Where is Dan from?

Context:

Dan is a professor at UT.

Dan grew up in Houston.

Dan lives in Austin now.

Answer:

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RAG

- How do we find the best context to put into our prompt?
- Semantic similarity
  - The weather is nice outside.
  - It's 75 degrees and partly cloudy today.
  - We are outside today partly because 75 nice clouded leopards are a few degrees north of here.
- We can get an embedding for each token in each block using an encoder transformer
  - How do we use those to determine similarity?

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Semantic Similarity

- Start with a BERT model
- Feed text-block-A and text-block-B into BERT
- Get contextually aware embeddings for each token in each block
- Average the embeddings in text-block-A
- Average the embeddings in text-block-B
- Calculate cosine similarity of 2 averages

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Semantic Similarity

- Sentences that have similar context will have high similarity
- This isn't perfect though
- Fine tune BERT
  - Dataset of similar blocks of text
    - Overflow repeat tagged questions
  - Dataset of question/answer pairs
    - ELI5
- [BOS] token
- There are many pre-trained models available

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Vector Database

- Break your tractor manual into lots of chunks
  - Most models only accept 512 tokens…
- Embed every chunk using a pretrained model
  - Store each embedding in a database
- FAISS / Pinecone / Chroma / Elastic / …

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RAG Software

- LangChain / LlamaIndex
- C3 AI
- Westlaw Precision
- Capstone

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LangChain

1. Create database
   a. Input directory of all documents
   b. Split each document into overlapping chunks
   c. Embed each chunk
   d. Save to database (FAISS)
2. Embed query
3. Find top results
4. Feed results and query to generative LLM

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Agentic RAG

- What if we have multiple databases to search?
- What if we have multiple search methodologies?

- Let an LLM decide!
  - Agentic RAG

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